Neural ODEs and Physics-Informed Learning for Interfacial Polymerization Kinetics in Thermoplastic Polyamide Elastomer Synthesis

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Introduction

Thermoplastic polyamide elastomers (TPAEs) consisting of poly(tetramethylene oxide) (PTMO) soft segments and Nylon-6,10 hard segments are emerging as sustainable alternatives to neoprene in wetsuit manufacturing [1]. Their synthesis via interfacial polymerization (IP) — a rapid, room-temperature process where a growing polymer film forms at the interface of two immiscible liquids — enables direct three-dimensional shaping without thermoforming. However, the reaction kinetics governing molecular weight, conversion, and final properties remain poorly understood for this copolymer system.

Scientific machine learning (SciML) offers two complementary tools for this problem. *Neural ODEs* [2] replace the known-but-approximate kinetic model with a learned dynamics function, capturing effects such as film diffusion resistance and PTMO-dependent reactivity that analytical models miss. *Physics-informed neural networks* (PINNs) [3] embed ODE residuals as a loss term, enabling recovery of unknown parameters such as the bimolecular rate constant k from sparse experimental measurements — exactly the scenario encountered during early-stage synthesis campaigns.

This project applies both approaches to IP kinetics for six TPAE compositions (5–60 wt% PTMO), connecting reaction outcomes to structure–property relationships via the Carothers and Flory–Fox equations.

Method

Part 1: ODE simulation. IP kinetics are modeled in the well-mixed approximation as a coupled second-order system:

$$\frac{d[A]}{dt} = -k[A][B], \quad \frac{d[B]}{dt} = -k[A][B] \tag{1}$$

where $[A]$ is the diamine (aqueous) and $[B]$ is the diacyl chloride (organic). The initial diamine concentration varies with composition, $[A]_0 = w_{\text{Nylon}}[A]_{0,N} + w_{\text{PTMO}}[A]_{0,P}$, reflecting the lower reactivity of PTMO ether-diamines. Eq. (1) is solved with RK45 and compared to forward Euler at three step sizes.

Part 2: Neural ODE. The RHS of Eq. (1) is replaced by an MLP $f_\theta([A],[B])$ (two hidden layers of 16 neurons, tanh activations). The network is trained on the 20% PTMO trajectory with added Gaussian noise ($\sigma = 0.005$ mol/L). Gradients are computed via backpropagation through unrolled Euler steps (BPTT), equivalent to the adjoint method [2]. Generalization is tested on three held-out compositions.

Part 3: PINN inverse problem. Given only $N = 15$ sparse noisy observations, we ask: can k be recovered? A PINN $u_\theta(t) \approx ([A](t),[B](t))$ is trained on:

$$\mathcal{L} = \underbrace{\frac{1}{N}\sum_i (u_\theta(t_i) - y_i)^2}_{\mathcal{L}_{\text{data}}} + \lambda \underbrace{\frac{1}{N_c}\sum_j \left(\left.\frac{du}{dt}\right|_{t_j} + k\, u_A u_B \right)^2}_{\mathcal{L}_{\text{physics}}} \tag{2}$$

where k is a *trainable parameter* initialized at $k_0 = 0.2$ L/mol/s (true $k = 0.8$). Physics residuals enforce Eq. (1) at $N_c = 100$ unlabeled collocation points.

Coda. Degree of polymerization X_n from Carothers feeds the Flory–Fox correction $T_g = T_{g,\infty} - K/M_n$, connecting kinetics to final thermal properties.

Results

Part 1. Figure 1 shows smooth monotonic conversion curves for all six compositions. Higher PTMO content reduces $[A]_0$, slightly increasing final conversion ($X_\infty \approx$ 0.37–0.41) as the diamine approaches stoichiometric balance with $[B]_0 = 0.08$ mol/L. Final X_n decreases from 18.2 to 9.1 as PTMO increases from 5% to 60%. Forward Euler at $\Delta t = 2.0$ s diverges visibly from RK45, confirming the need for adaptive step-size control.

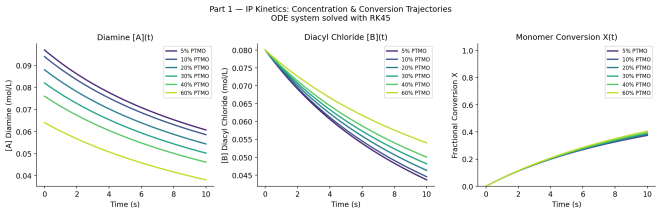

Figure 1: Part 1 — Diamine [A](t), diacyl chloride [B](t), and fractional conversion X(t) for all six TPAE compositions solved with RK45. Each line is one PTMO weight fraction (5–60%).

Part 2. The Neural ODE fits the 20% PTMO training trajectory with MAE $< 10^{-3}$ mol/L after 600 epochs (Figure 2). Generalization to unseen compositions achieves MAE $< 4 \times 10^{-3}$ mol/L for $[A]$, confirming that the learned dynamics transfer across initial conditions. The learned rate surface qualitatively recovers the bilinear structure of $k[A][B]$, but with systematic overestimation in the high-concentration regime not visited during training.

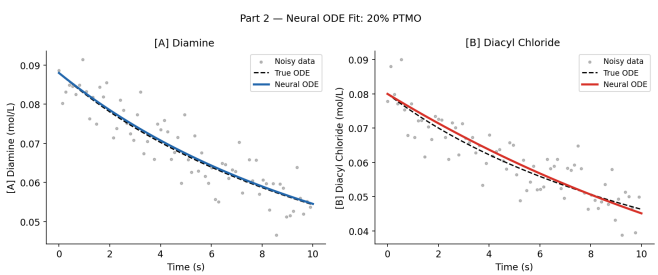

Figure 2: Part 2 — Neural ODE fit on the 20% PTMO training composition. Grey dots: noisy observations. Dashed: true ODE. Solid: learned trajectory. The network infers d[A]/dt from data without being given the $k[A][B]$ form.

Part 3. Starting from $k_0 = 0.2$ L/mol/s, the PINN converges monotonically to $k = 0.72$ L/mol/s over 3000 epochs (9.9% error; Figure 3). The physics loss drives k upward: without it, the data loss alone cannot distinguish the correct k from nearby values over the narrow observed concentration range. A λ sweep confirms moderate weighting ($\lambda = 5$) is optimal; over-constraining ($\lambda = 20$) degrades recovery.

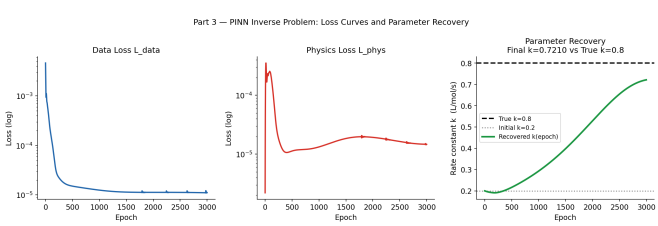

Figure 3: Part 3 — PINN inverse problem. Left: data loss. Center: physics loss (early spike reflects tension between fitting data and satisfying the ODE). Right: k converges from 0.2 toward $k_{\text{true}} = 0.8$ L/mol/s using only 15 sparse observations.

Coda. X_n ranges from 9.1 (60% PTMO) to 18.2 (5% PTMO), yielding $M_n = 2200$–4400 g/mol. The Flory–Fox correction depresses T_g by up to 30 °C below the Fox blending prediction, a direct consequence of the kinetics-limited chain length. Modulus spans three decades (63–2040 MPa) across the composition range.

Discussion

The three-part study demonstrates a natural progression from classical ODE modeling to data-driven and physics-informed ML, motivated by the experimental realities of IP synthesis. The Neural ODE result highlights a fundamental limitation of single-trajectory training: insufficient coverage of concentration space leads to systematic bias under extrapolation. Training on multiple compositions simultaneously — feasible once UV-VIS kinetic data become available — would resolve this.

The PINN recovers k to within 10% from 15 observations starting from a 4× wrong guess. The residual error reflects genuine identifiability limits: over the observed concentration range, many k values produce comparable trajectory fits. Bayesian methods (Monte Carlo dropout, Gaussian process priors) would quantify this uncertainty, connecting to recent statistical analysis of PINNs [5].

Most importantly, the coda shows that kinetics are not merely a means to an end: X_n determined by the IP conditions modulates T_g via Flory–Fox by up to 30 °C. Controlling synthesis conditions to maximize X_n is therefore a direct lever for targeting thermal properties in wetsuit applications.

References

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